

Technical Summary:

Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs

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1 Research Problem and Motivation

Large Language Models (LLMs) are frequently used through black-box APIs, where internal model signals such as logits and hidden states are inaccessible. In such settings, downstream systems often still require a confidence score to decide when to trust an answer, when to request human review, or when to trigger fallback tools. The core research problem is *confidence elicitation*: estimating an LLM’s confidence in its response without model fine-tuning or internal access.

A central motivation is that directly using token-level likelihood as a proxy for confidence is not always aligned with the semantic correctness of the answer. The study argues that *verbalized confidence* (a model explicitly stating a confidence value for its produced answer) is intrinsically tied to the meaning of the answer rather than lexical form, and therefore forms a plausible black-box alternative. However, empirical evidence indicates that vanilla verbalized confidence is frequently overconfident and exhibits limited capability for separating correct from incorrect outputs, especially on more complex tasks. A systematic evaluation across models, tasks, and elicitation strategies is therefore necessary to identify which black-box methods improve (i) calibration and (ii) failure prediction, and to characterize remaining bottlenecks.

2 Related Work

Prior work on confidence elicitation for LLMs includes prompting models to output verbalized confidence, fine-tuning-based approaches that require confidence supervision, and external calibrators that rely on internal representations that are unavailable in black-box deployments. Additional studies analyze confidence behaviors or present prompting-only strategies such as Top- K style elicitation, but the explored method space is relatively narrow.

The study positions itself as an empirical synthesis that (i) defines confidence elicitation in black-box settings, (ii) explores a broader combination space via a unified framework, and (iii) evaluates a wider range of models beyond a single family of RLHF-tuned systems. A key comparative point is that white-box, token-probability-based methods often perform better, but reported gaps in failure prediction can be narrow, motivating continued investment in black-box strategies that remain deployable under API constraints.

3 Dataset Construction

No new dataset is constructed. The evaluation uses established benchmarks spanning commonsense reasoning, arithmetic reasoning, symbolic reasoning, professional knowledge, and ethical knowledge. Datasets are drawn from sources including BigBench and MMLU, and representative datasets are selected within each task family for primary analyses.

4 Query Protocol and Task Definitions

4.1 Black-box confidence elicitation protocol

Each evaluation instance consists of an input question and a model output that includes (i) an answer and (ii) a numerical confidence in the range 0–100. Prompt formats vary by question type (open-number versus multiple-choice) and by the prompting strategy. Across prompts, an explicit instruction is appended to clarify the meaning of confidence as the likelihood the answer is true.

4.2 Tasks and success criteria

The study evaluates confidence quality under two task definitions:

- **Confidence calibration:** confidence values should match empirical correctness rates. The primary metric is Expected Calibration Error (ECE), described as the discrepancy between predicted probabilities and observed accuracy. Lower ECE indicates better calibration.
- **Failure prediction:** confidence scores should discriminate between correct and incorrect answers. The primary metric is AUROC, which measures discrimination across thresholds. AUROC near 0.5 corresponds to random guessing. The study also reports AUPRC for positive and negative classes to characterize precision-recall trade-offs.

5 Modeling Approach

5.1 Unified framework

Black-box confidence elicitation methods are expressed as combinations of three components: **prompting strategy** (how input-output format elicits confidence), **sampling strategy** (how many responses are generated and under what variations), and **aggregation strategy** (how sampled outputs are combined into a final answer and confidence). Given an input question, a prompt strategy is chosen, M responses are sampled, and an aggregator computes confidences for candidate answers; the highest-confidence answer is returned.

5.2 Prompting strategies

The evaluated prompting strategies include Vanilla, Chain-of-Thought (CoT), Self-Probing, Multi-Step, and Top- K . CoT is treated as a reasoning-augmented strategy that often improves accuracy; Self-Probing separates answer generation and confidence elicitation into independent sessions; Multi-Step requests per-step confidence; Top- K requests multiple guesses with probabilities.

5.3 Aggregation strategies and core formulas

Let $\hat{a}^{(i)}$ denote the answer in the i -th sampled response and $\hat{c}^{(i)} \in [0, 1]$ denote its verbalized confidence scaled from 0–100 to 0–1. Let A be the set of unique answers among the M samples.

Consistency aggregation.

$$C_{\text{cons}}(a) = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[\hat{a}^{(i)} = a]. \quad (1)$$

Purpose: estimate confidence from answer agreement across samples, primarily to support failure prediction by leveraging response variance.

Symbols: M is the number of sampled responses; $\hat{a}^{(i)}$ is the i -th sampled answer; $\mathbf{1}[\cdot]$ is an indicator function; $a \in A$ is a candidate answer.

Intuition: an answer that appears frequently across samples is treated as more reliable.

Usage: after sampling M responses, compute $C_{\text{cons}}(a)$ for each $a \in A$ and select the answer with the largest value.

Average-confidence aggregation.

$$C_{\text{avg}}(a) = \frac{\sum_{i=1}^M \hat{c}^{(i)} \mathbf{1}[\hat{a}^{(i)} = a]}{\sum_{i=1}^M \mathbf{1}[\hat{a}^{(i)} = a]}. \quad (2)$$

Purpose: combine verbalized confidence with answer identity to yield a finer-grained confidence estimate for calibration.

Symbols: $\hat{c}^{(i)} \in [0, 1]$ is the verbalized confidence in sample i ; other symbols follow Equation (1).

Intuition: among samples that propose the same answer, the average stated confidence is taken as the confidence of that answer.

Usage: after sampling, compute $C_{\text{avg}}(a)$ for each candidate a and return the argmax as the final output with its confidence.

Multi-step confidence aggregation. For a prompt that decomposes reasoning into n steps with per-step confidence values $C_i \in [0, 1]$, the overall confidence is defined as:

$$C_{\text{multi-step}} = \prod_{i=1}^n C_i. \quad (3)$$

Purpose: penalize final confidence when any intermediate reasoning step is assigned low confidence, targeting overconfidence reduction.

Symbols: C_i is the confidence assigned to step i ; n is the number of steps.

Intuition: correctness of the final answer requires correctness of all steps, so confidence decreases multiplicatively when uncertainty exists in any step.

Usage: Multi-Step prompting elicits stepwise confidences and computes $C_{\text{multi-step}}$ as the final confidence reported for the final answer.

5.4 Minimal example of the framework

Consider an open-number question where three sampled responses produce answers 20, 25, and 100 with confidences 0.90, 0.80, and 0.37, respectively. With Average-confidence aggregation (Equation (2)), each answer appears once, so its aggregated confidence equals its stated confidence. The framework outputs answer 20 with confidence 0.90. This illustrates how the framework selects among candidate answers after sampling and aggregation.

6 Empirical Results

6.1 Vanilla verbalized confidence is poorly calibrated and weak for failure prediction

Across evaluated models and datasets, vanilla verbalized confidence exhibits substantial miscalibration, with average ECE reported above 0.377 for GPT-3, GPT-3.5, and Vicuna. GPT-4 reduces average ECE to 0.18 and achieves an average AUROC of 0.627, representing a reported AUROC improvement of 22.2% relative to GPT-3. Despite this improvement, AUROC remains far from perfect discrimination, indicating persistent limitations in ranking incorrect versus correct outputs using confidence alone.

6.2 Prompting strategies can reduce ECE but do not guarantee better discrimination

Prompting strategies such as CoT, Multi-Step, and Top- K reduce ECE and can improve AUROC on average, with Top- K described as relatively more effective. However, the study highlights a failure mode where ECE improvements arise from increased task accuracy and a narrower gap between mean confidence and accuracy, rather than improved ability to distinguish correct from incorrect samples. A concrete example is CoT on GSM8K with GPT-4: accuracy reaches 93.6% and ECE is 0.064 by assigning 100% confidence to all samples, which provides no ranking signal for failure prediction because confidence is constant.

6.3 Sampling multiple responses substantially improves failure prediction

Variance among multiple sampled responses provides a strong signal for failure prediction. Table 2 reports sampling strategies on GPT-3.5 with $M = 5$, where CoT and Consistency are used for $M > 1$. Self-Random sampling achieves AUROC 0.927 on GSM8K versus AUROC 0.548 for CoT with $M = 1$, and improves average AUROC to 0.730 versus 0.564 for CoT with $M = 1$. This demonstrates that repeated sampling and consistency-based aggregation can transform confidence into a useful failure predictor, particularly on arithmetic reasoning.

6.4 Aggregation strategies trade off calibration and failure prediction

Aggregation strategies differ in which metric they optimize. Consistency yields coarse-grained confidence values (for $M = 5$, limited to six possible values), which can harm calibration. Aggregation using verbalized confidences (such as Average-confidence and Pair-Rank) provides finer granularity and improves calibration, especially when query budgets are constrained. A reported result is that Pair-Rank reduces ECE to 0.028 on DateUnd and achieves mean ECE 0.069 across five datasets under a fixed setting, while maintaining competitive AUROC. These findings support selecting aggregators based on deployment priorities (calibration versus failure prediction) and query cost constraints.

6.5 Comparison with white-box baselines

Token-probability-based white-box methods (sequence probability, length-normalized sequence probability, and key token probability) are evaluated where available. White-box methods often outperform black-box verbalized confidence, but the reported AUROC gap is narrow in some comparisons (0.522 to 0.605). This result indicates that black-box elicitation remains competitive in certain settings while retaining deployability under API-only constraints.

7 Summary

7.1 Key contributions and innovations relative to prior work

- **Unified black-box framework:** confidence elicitation methods are organized into a combinatorial framework with prompt, sampling, and aggregation components, enabling systematic exploration beyond prompting-only methods.
- **Broader evaluation scope:** strategies are empirically compared across multiple task families and across multiple model families, rather than focusing on a single model type or a single prompting format.

- **Mechanistic insights on what improves what:** prompting strategies can reduce ECE primarily through accuracy gains, while multi-sample variance materially improves failure prediction; aggregation choice induces explicit trade-offs between calibration and discrimination.

7.2 Limitations and future directions grounded in reported results

- **Failure prediction remains challenging on specialized tasks:** the study notes persistent difficulty in failure prediction, especially for tasks requiring specialized knowledge, and emphasizes the need for further research.
- **Overconfidence patterns remain:** confidence outputs concentrate in high ranges (notably 80% to 100% and multiples of 5), limiting usefulness for fine-grained discrimination even when calibration improves.
- **Compute and query-cost constraints:** sampling-based improvements require multiple queries per instance, and computational constraints limit M , which restricts how well sampling approximates the model’s answer distribution.

Table 1: Evaluation benchmarks and selection rationale. Dataset sizes, preprocessing details, and split specifications are not provided in the paper.

Task Family	Dataset	Source	Notes (selection and characteristics)
Commonsense Reasoning	SportUND	BigBench	Included; StrategyQA selected as representative due to containing more data.
Commonsense Reasoning	StrategyQA	BigBench	Selected as representative dataset for commonsense reasoning.
Arithmetic Reasoning	GSM8K	Cobbe et al. (2021)	Selected as representative due to wider usage.
Arithmetic Reasoning	SVAMP	Patel et al. (2021)	Included as an arithmetic benchmark.
Symbolic Reasoning	DateUnd	BigBench	Selected as representative due to being more difficult than ObjectCou.
Symbolic Reasoning	ObjectCou	BigBench	Included as a symbolic reasoning benchmark.
Professional Knowledge	Prf-Law	MMLU	Domain-specific multiple-choice questions.
Ethical Knowledge	Biz-Ethics	MMLU	Domain-specific multiple-choice questions.
Scale / Splits	Not specified in the paper.	Not specified in the paper.	Not specified in the paper.

Table 2: Sampling strategies on GPT-3.5 with $M = 5$ (metrics scaled by 10^2 in the paper). Lower ECE is better; higher AUROC is better.

Method	GSM8K (ECE/AUROC)	Prf-Law (ECE/AUROC)	DateUnd (ECE/AUROC)	StrategyQA (ECE/AUROC)	Biz-Ethics (ECE/AUROC)	Avg (ECE/AUROC)
Misleading ($M = 5$)	8.03 / 88.6	18.3 / 59.3	20.5 / 67.3	21.8 / 61.5	17.8 / 71.3	17.3 / 69.6
Self-Random ($M = 5$)	6.28 / 92.7	26.0 / 65.6	17.0 / 66.8	23.3 / 60.8	20.7 / 79.0	18.7 / 73.0
Prompt ($M = 5$)	35.2 / 74.4	31.5 / 60.8	23.9 / 69.8	16.1 / 61.3	15.0 / 79.5	24.3 / 69.2
CoT ($M = 1$)	10.1 / 54.8	39.7 / 52.2	23.4 / 57.4	22.0 / 59.8	30.0 / 56.0	25.0 / 56.4
Top- K ($M = 1$)	19.6 / 58.5	16.7 / 58.9	26.1 / 74.2	14.0 / 61.3	12.4 / 73.3	17.8 / 65.2